



MATH130: Exponential and Logarithmic Functions

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Learning Objectives

- Evaluate and graph exponential functions
- Convert between exponential and logarithmic form
- Apply properties of logarithms to simplify and solve
- Solve exponential equations using logarithms

Simplify each expression completely. Show all steps and circle your final answer.

Evaluating exponential expressions

1. Evaluate: 2^2 .

$$2^2$$

Answer: _____

2. Evaluate: 2^5 .

$$2^5$$

Answer: _____

3. Evaluate: 5^4 .

$$5^4$$

Answer: _____

4. Evaluate: 3^5 .

$$3^5$$

Answer: _____

5. Evaluate: 3^3 .

$$3^3$$

Answer: _____

6. Evaluate: 3^2 .

$$3^2$$

Answer: _____

7. Evaluate: 2^4 .

$$2^4$$

Answer: _____

8. Evaluate: 2^4 .

$$2^4$$

Answer: _____

Solving exponential equations

9. Solve for x: $2^x = 8$. Express the answer as an integer.

$$\log_2 8$$

Answer: _____

10. Solve for x: $3^x = 3$. Express the answer as an integer.

$$\log_3 3$$

Answer: _____

11. Solve for x: $2^x = 32$. Express the answer as an integer.

$$\log_2 32$$

Answer: _____

12. Solve for x: $2^x = 4$. Express the answer as an integer.

$$\log_2 4$$

Answer: _____

13. Solve for x: $2^x = 2$. Express the answer as an integer.

$$\log_2 2$$

Answer: _____

14. Solve for x: $3^x = 81$. Express the answer as an integer.

$$\log_3 81$$

Answer: _____

15. Solve for x: $2^x = 2$. Express the answer as an integer.

$$\log_2 2$$

Answer: _____

16. Solve for x: $3^x = 243$. Express the answer as an integer.

$$\log_3 243$$

Answer: _____

Evaluating logarithms

17. Evaluate: $\log_{10}(10)$.

$$\log_{10} 10$$

Answer: _____

18. Evaluate: $\log_5(25)$.

$$\log_5 25$$

Answer: _____

19. Evaluate: $\log_5(5)$.

$$\log_5 5$$

Answer: _____

20. Evaluate: $\log_2(16)$.

$$\log_2 16$$

Answer: _____

21. Evaluate: $\log_5(125)$.

$$\log_5 125$$

Answer: _____

22. Evaluate: $\log_{10}(10)$.

$$\log_{10} 10$$

Answer: _____

23. Evaluate: $\log_{10}(10)$.

$$\log_{10} 10$$

Answer: _____

Properties of logarithms

24. Condense to a single logarithm: $\log_3(9) + \log_3(9)$.

$$\log_3 9 + \log_3 9$$

Answer: _____

25. Condense to a single logarithm: $\log_2(4) + \log_2(4)$.

$$\log_2 4 + \log_2 4$$

Answer: _____

26. Condense to a single logarithm: $\log_{10}(10) + \log_{10}(100)$.

$$\log_{10} 10 + \log_{10} 100$$

Answer: _____

27. Condense to a single logarithm: $\log_2(4) + \log_2(8)$.

$$\log_2 4 + \log_2 8$$

Answer: _____

28. Condense to a single logarithm: $\log_3(3) + \log_3(27)$.

$$\log_3 3 + \log_3 27$$

Answer: _____

29. Condense to a single logarithm: $\log_3(9) + \log_3(9)$.

$$\log_3 9 + \log_3 9$$

Answer: _____

30. Condense to a single logarithm: $\log_{10}(10) + \log_{10}(10)$.

$$\log_{10} 10 + \log_{10} 10$$

Answer: _____



Topics: Properties of logarithms, Solving exponential equations, Evaluating exponential expressions, Evaluating logarithms. All answers verified by independent computation.

Solutions

Evaluating exponential expressions

1. Evaluate: 2^2 .

$$2^2$$

$$\rightarrow 2^2 = 4.$$

Answer: $2^2 = 4$

2. Evaluate: 2^5 .

$$2^5$$

$$\rightarrow 2^5 = 32.$$

Answer: $2^5 = 32$

3. Evaluate: 5^4 .

$$5^4$$

$$\rightarrow 5^4 = 625.$$

Answer: $5^4 = 625$

4. Evaluate: 3^5 .

$$3^5$$

$$\rightarrow 3^5 = 243.$$

Answer: $3^5 = 243$

5. Evaluate: 3^3 .

$$3^3$$

$$\rightarrow 3^3 = 27.$$

Answer: $3^3 = 27$

6. Evaluate: 3^2 .

$$3^2$$

$$\rightarrow 3^2 = 9.$$

Answer: $3^2 = 9$

7. Evaluate: 2^4 .

$$2^4$$

$$\rightarrow 2^4 = 16.$$

Answer: $2^4 = 16$

8. Evaluate: 2^4 .

$$2^4$$

$$\rightarrow 2^4 = 16.$$

Answer: $2^4 = 16$

Solving exponential equations

9. Solve for x : $2^x = 8$. Express the answer as an integer.

$$\log_2 8$$

→ Rewrite: $x = \log_2(8)$. Since $2^3 = 8$, $x = 3$.

Answer: $\log_2 8 = 3$

10. Solve for x : $3^x = 3$. Express the answer as an integer.

$$\log_3 3$$

→ Rewrite: $x = \log_3(3)$. Since $3^1 = 3$, $x = 1$.

Answer: $\log_3 3 = 1$

11. Solve for x : $2^x = 32$. Express the answer as an integer.

$$\log_2 32$$

→ Rewrite: $x = \log_2(32)$. Since $2^5 = 32$, $x = 5$.

Answer: $\log_2 32 = 5$

12. Solve for x : $2^x = 4$. Express the answer as an integer.

$$\log_2 4$$

→ Rewrite: $x = \log_2(4)$. Since $2^2 = 4$, $x = 2$.

Answer: $\log_2 4 = 2$

13. Solve for x : $2^x = 2$. Express the answer as an integer.

$$\log_2 2$$

→ Rewrite: $x = \log_2(2)$. Since $2^1 = 2$, $x = 1$.

Answer: $\log_2 2 = 1$

14. Solve for x : $3^x = 81$. Express the answer as an integer.

$$\log_3 81$$

→ Rewrite: $x = \log_3(81)$. Since $3^4 = 81$, $x = 4$.

Answer: $\log_3 81 = 4$

15. Solve for x : $2^x = 2$. Express the answer as an integer.

$$\log_2 2$$

→ Rewrite: $x = \log_2(2)$. Since $2^1 = 2$, $x = 1$.

Answer: $\log_2 2 = 1$

16. Solve for x : $3^x = 243$. Express the answer as an integer.

$$\log_3 243$$

→ Rewrite: $x = \log_3(243)$. Since $3^5 = 243$, $x = 5$.

Answer: $\log_3 243 = 5$

Evaluating logarithms

17. Evaluate: $\log_{10}(10)$.

$$\log_{10} 10$$

→ Ask: $10^? = 10$. Since $10^1 = 10$, $\log_{10}(10) = 1$.

Answer: $\log_{10} 10 = 1$

18. Evaluate: $\log_5(25)$.

$$\log_5 25$$

→ Ask: $5^? = 25$. Since $5^2 = 25$, $\log_5(25) = 2$.

Answer: $\log_5 25 = 2$

19. Evaluate: $\log_5(5)$.

$$\log_5 5$$

→ Ask: $5^? = 5$. Since $5^1 = 5$, $\log_5(5) = 1$.

Answer: $\log_5 5 = 1$

20. Evaluate: $\log_2(16)$.

$$\log_2 16$$

→ Ask: $2^? = 16$. Since $2^4 = 16$, $\log_2(16) = 4$.

Answer: $\log_2 16 = 4$

21. Evaluate: $\log_5(125)$.

$$\log_5 125$$

→ Ask: $5^? = 125$. Since $5^3 = 125$, $\log_5(125) = 3$.

Answer: $\log_5 125 = 3$

22. Evaluate: $\log_{10}(10)$.

$$\log_{10} 10$$

→ Ask: $10^? = 10$. Since $10^1 = 10$, $\log_{10}(10) = 1$.

Answer: $\log_{10} 10 = 1$

23. Evaluate: $\log_{10}(10)$.

$$\log_{10} 10$$

→ Ask: $10^? = 10$. Since $10^1 = 10$, $\log_{10}(10) = 1$.

Answer: $\log_{10} 10 = 1$

Properties of logarithms

24. Condense to a single logarithm: $\log_3 9 + \log_3 9$.

$$\log_3 9 + \log_3 9$$

→ Product rule: $\log_b(A) + \log_b(B) = \log_b(A \cdot B)$.

$$\rightarrow \log_3 9 + \log_3 9 = \log_3(81) = 4.$$

Answer: $\log_3 81 = 4$

25. Condense to a single logarithm: $\log_2 4 + \log_2 4$.

$$\log_2 4 + \log_2 4$$

→ Product rule: $\log_b(A) + \log_b(B) = \log_b(A \cdot B)$.

$$\rightarrow \log_2 4 + \log_2 4 = \log_2(16) = 4.$$

Answer: $\log_2 16 = 4$

26. Condense to a single logarithm: $\log_{10} 10 + \log_{10} 100$.

$$\log_{10} 10 + \log_{10} 100$$

→ Product rule: $\log_b(A) + \log_b(B) = \log_b(A \cdot B)$.

$$\rightarrow \log_{10} 10 + \log_{10} 100 = \log_{10}(1000) = 3.$$

Answer: $\log_{10} 1000 = 3$

27. Condense to a single logarithm: $\log_2 4 + \log_2 8$.

$$\log_2 4 + \log_2 8$$

→ Product rule: $\log_b(A) + \log_b(B) = \log_b(A \cdot B)$.

$$\rightarrow \log_2 4 + \log_2 8 = \log_2(32) = 5.$$

Answer: $\log_2 32 = 5$

28. Condense to a single logarithm: $\log_3 3 + \log_3 27$.

$$\log_3 3 + \log_3 27$$

→ Product rule: $\log_b(A) + \log_b(B) = \log_b(A \cdot B)$.

$$\rightarrow \log_3 3 + \log_3 27 = \log_3(81) = 4.$$

Answer: $\log_3 81 = 4$

29. Condense to a single logarithm: $\log_3 9 + \log_3 9$.

$$\log_3 9 + \log_3 9$$

→ Product rule: $\log_b(A) + \log_b(B) = \log_b(A \cdot B)$.

$$\rightarrow \log_3 9 + \log_3 9 = \log_3(81) = 4.$$

Answer: $\log_3 81 = 4$

30. Condense to a single logarithm: $\log_{10}(10) + \log_{10}(10)$.

$$\log_{10} 10 + \log_{10} 10$$

→ *Product rule:* $\log_b(A) + \log_b(B) = \log_b(A*B)$.

→ $\log_{10}(10) + \log_{10}(10) = \log_{10}(100) = 2$.

Answer: $\log_{10} 100 = 2$
